Approximate feasibility can miss the intended constraint

One pair, one criterion; M = 1.
Exact forward error = t + 1/10;
approximate forward error = t.

t = 0; q = 1/20; ε = 1/10

q − ε = -1/20; q + ε = 3/20

Report	N	$\widehat{E}$	E
Hold	0	0	0
A>B	1	0	1/10
B>A	1	1	9/10

Report	Inner	Exact	Nominal	Outer
Hold	**Yes**	**Yes**	**Yes**	**Yes**
A>B	No	No	**Yes**	**Yes**
B>A	No	No	No	No

t = 1/5; q = 1/4; ε = 1/10

q − ε = 3/20; q + ε = 7/20

Report	N	$\widehat{E}$	E
Hold	0	0	0
A>B	1	1/5	3/10
B>A	1	4/5	7/10

Report	Inner	Exact	Nominal	Outer
Hold	**Yes**	**Yes**	**Yes**	**Yes**
A>B	No	No	**Yes**	**Yes**
B>A	No	No	No	No

Inner: $\widehat{E} \le (q-\varepsilon)N$
Exact: $E \le qN$
Nominal: $\widehat{E} \le qN$
Outer: $\widehat{E} \le (q+\varepsilon)N$

At t = 0, A>B has N = 1.
Inner 0 ≤ −1/20 fails.
Clipping to 0 wrongly passes;
exact 1/10 > q = 1/20 still fails.

Hold: N = 0; all count rows read 0 ≤ 0.
Its error fraction is undefined.
Exact refers to the intended supplied law.